Bug Report Classifier - Requirements

Jack Honour [3063159]

Hardware Requirements

Tested Systems

System 1 (Desktop)

- OS: Windows 10 Pro (64-bit)
- CPU: AMD Ryzen 9 7950X3D (16 cores, 32 threads @ 5.2GHz)
- RAM: 64GB (6000 MT/s)
- GPU: NVIDIA RTX 4070 Ti Super (16GB VRAM)

System 2 (Laptop)

- OS: macOS Tahoe 26.4.1 (ARM64)
- CPU: Apple M1 (8 cores)
- RAM: 8GB (Unified Memory)
- GPU: Apple M1 GPU (8 cores)

System 3 (Linux Laptop)

- OS: Fedora 43 (Asahi - ARM64)
- CPU: Apple M1 (8 cores)
- RAM: 8GB (Unified Memory)
- GPU: Apple M1 GPU (8 cores)

Recommended Requirements

- **Operating System:** Windows 10/11 (64-bit), macOS (64-bit), Linux (64-bit)
- **CPU:** $\geq$ Quad-core (e.g., Intel Core i5 / AMD Ryzen 5)
- **RAM:** $\geq$ 16GB
- **Storage:** $\geq$ 7GB free space (for models)

Linux users should be able to run the code on similar hardware (GPU not required), but may need to adjust installation steps due to PyTorch compatibility issues.

Software Requirements

- Python 3.12.8 (https://www.python.org/downloads/release/python-3128/)
- Git (https://git-scm.com)
- Git LFS (https://git-lfs.com)

Python Dependencies

It is recommended to use a virtual environment (e.g., venv, conda) to manage Python dependencies and avoid conflicts with system packages.

```
python --version  # Ensure Python 3.12.x is installed and currently active
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
```

Required packages are locked and listed in `requirements.txt` and can be installed via pip:

```
(venv) pip install -r requirements.txt
```

requirements.txt

```
--extra-index-url https://download.pytorch.org/whl/cpu
accelerate==1.13.0
annotated-doc==0.0.4
anyio==4.13.0
certifi==2026.4.22
click==8.3.3
colorama==0.4.6
contourpy==1.3.3
cycler==0.12.1
filelock==3.29.0
fonttools==4.62.1
fsspec==2026.3.0
gensim==4.4.0
h11==0.16.0
hf-xet==1.4.3
httpcore==1.0.9
httpx==0.28.1
huggingface_hub==1.11.0
```

```
idna==3.13
Jinja2==3.1.6
joblib==1.5.3
kiwisolver==1.5.0
markdown-it-py==4.0.0
MarkupSafe==3.0.3
matplotlib==3.10.8
mdurl==0.1.2
mpmath==1.3.0
networkx==3.6.1
nltk==3.9.4
numpy==2.4.4
packaging==26.1
pandas==3.0.2
pillow==12.2.0
psutil==7.2.2
Pygments==2.20.0
pyparsing==3.3.2
python-dateutil==2.9.0.post0
PyYAML==6.0.3
regex==2026.4.4
rich==15.0.0
safetensors==0.7.0
scikit-learn==1.8.0
scipy==1.17.1
seaborn==0.13.2
setuptools==81.0.0
shellingham==1.5.4
six==1.17.0
smart_open==7.6.0
sympy==1.14.0
threadpoolctl==3.6.0
tokenizers==0.22.2
torch==2.11.0+cpu
torchvision==0.26.0+cpu
tqdm==4.67.3
transformers==5.6.2
typer==0.24.2
typing_extensions==4.15.0
tzdata==2026.1
wrapt==2.1.2
```

Linux OS Notes

While these steps worked ok on my machines, PyTorch may not install nicely on Linux.

Please please do not mark down due to this, this is completely out of my control. The code should run without GPU support, albeit much slower.

I have specifically not used the GPU to make this as compatible as possible, please visit `https://pytorch.org/get-started/locally/` if the requirement cannot automatically be met.